

MUNICIPAL BUREAUCRACY & PERMITTING NAVIGATOR

VERTEXA

CIVIC-TASK-NAVIGATOR

Technical Architecture, Full Codebase Audit & Hackathon Master Report
Maharashtra Unified Development Control & Promotion Regulations (UDCPR 2020) & MRTP Act 1966

PROJECT TYPOLOGIES

Residential, Commercial, Institutional,
Hospitality, Mixed-Use, Industrial

FRONTEND STACK

React 19, Tailwind CSS v4, Vite 8,
@xyflow/react

DUAL ENGINE CORE

Deterministic UDCPR Rules + Gemini 3.5
Flash-Lite

BACKEND & SECURITY

Node 24, Express 5, SSRF Shield, Private
Subnet Filters

1. Executive Project Overview

Vertexa (Civic-Task-Navigator) is an intelligent, dual-engine civic platform designed to demystify, structure, and navigate the complex, multi-departmental building permission and statutory clearance ecosystem across Maharashtra's Urban Local Bodies (ULBs).

In Indian municipal governance—specifically governed by the **Maharashtra Regional and Town Planning (MRTP) Act, 1966** and the **Unified Development Control and Promotion Regulations (UDCPR 2020)**—securing permission to build a residential house, commercial office, school, or factory requires navigating up to 15 municipal departments, 25+ statutory certificates, and strict sequential gating milestones (such as IOD, CC, Plinth Verification, and OC).

The Vertexa Engineering Philosophy

Deterministic Mathematical Rules = Statutory Authority (100% legal accuracy, no AI hallucinations in legal conditions).

Google Gemini (gemini-3.5-flash-lite) = Plain-Language Assistance (Contextual breakdown of Marathi revenue terms and statutory bylaws).

Executive Pitch Formats

Vertexa in One Sentence:

"Vertexa is an intelligent civic navigation system that combines a deterministic UDCPR 2020 statutory rules engine with an interactive visual DAG roadmap and AI explanations to guide citizens through Maharashtra's building permission lifecycle."

Vertexa in 30 Seconds:

"Applying for building permissions in Maharashtra involves 5 sequential stages, dozens of municipal departments, and rigid statutory prerequisites under UDCPR 2020. Citizens are often stuck for months because one missing departmental NOC halts their Commencement Certificate. Vertexa replaces this bureaucratic nightmare with an interactive visual roadmap. Users input their plot parameters, our deterministic engine calculates exactly which clearances apply, our graph engine computes the dependency pipeline, and our AI explainer demystifies statutory jargon into plain English. It turns months of confusion into a structured, step-by-step master dossier."

Vertexa in 2 Minutes:

"Every year, hundreds of thousands of construction projects in Maharashtra—from simple G+2 residential homes in Pune to commercial malls in Mumbai—get entangled in municipal bureaucracy. Under the MRTP Act 1966 and UDCPR 2020, obtaining an Occupancy Certificate requires passing through 5 strict stages: Land Title Demarcation, PreDCR CAD Scrutiny, Conditional Sanction (IOD), Parallel Departmental NOCs (Fire, Tree, Drainage, Aviation, Heritage), Groundbreaking to Plinth Checking, and Superstructure to OC."

*Ordinary citizens lack access to town planning lawyers, and general-purpose LLMs hallucinate municipal bylaws and invent non-existent rules. Vertexa takes a principled engineering approach: **Deterministic Rules for Statutory Authority, AI for Plain-Language Explanation.***

When a user enters their requirement, Vertexa evaluates plot parameters through our codified UDCPR rules engine. It explicitly categorizes clearances into APPLIES, EXEMPT, DOES_NOT_APPLY, and REQUIRES_VERIFICATION. It then renders an interactive Directed Acyclic Graph (DAG) that enforces real-world

prerequisites: you cannot cast superstructure slabs before plinth verification, and you cannot obtain a Commencement Certificate before satisfying IOD clauses.

On top of this, Vertexa features a SSRF-protected live government link validator, an offline-resilient statutory fallback architecture, a Master Dossier kit for one-click printing of all required forms, and a Civic Jargon Buster powered by Gemini. It transforms opaque civic bureaucracy into a transparent, self-serve civic utility."

2. End-to-End User Workflow & Journey

- 1

Home Landing Screen & Custom Requirement (HomePage.jsx)
Citizen enters freeform requirement (e.g., "I want to construct a commercial shopping complex in Pune") or selects a preset.
- 2

Scope & Intent Classification Gate (scopeClassifier.js)
Evaluates intent: IN_SCOPE (preloads typology & city hints), OUT_OF_SCOPE (shows inline guidance banner), or NEEDS_CLARIFICATION.
- 3

Plot Questionnaire Intake (PlotQuestionnaireModal.jsx)
Collects building height, plot area, abutting road width, affected trees, airport funnel, heritage, ESZ, and HT power lines.
- 4

Deterministic Rules Evaluation (eligibilityEngine.js)
Evaluates 12+ codified UDCPR rules; categorizes into APPLIES, EXEMPT, DOES_NOT_APPLY, and REQUIRES_VERIFICATION.
- 5

API Pipeline Synthesis & Validation (POST /api/navigate)
Queries Gemini 3.5 Flash-Lite with grounded constraints & canonical IDs; validates via graphValidator.js with seed fallback.
- 6

Interactive React Flow Canvas (RoadmapCanvas.jsx & CivicNode.jsx)
Renders 5 stages of municipal permissions with locked/unlocked prerequisite gating and sequential auto-advance.
- 7

Master Dossier & Civic Jargon Buster (DocumentDrawer.jsx & JargonBusterModal.jsx)
Aggregates forms checklist with print view; provides real-time Gemini plain-language explanations of statutory jargon.

Construction Typologies Architecture

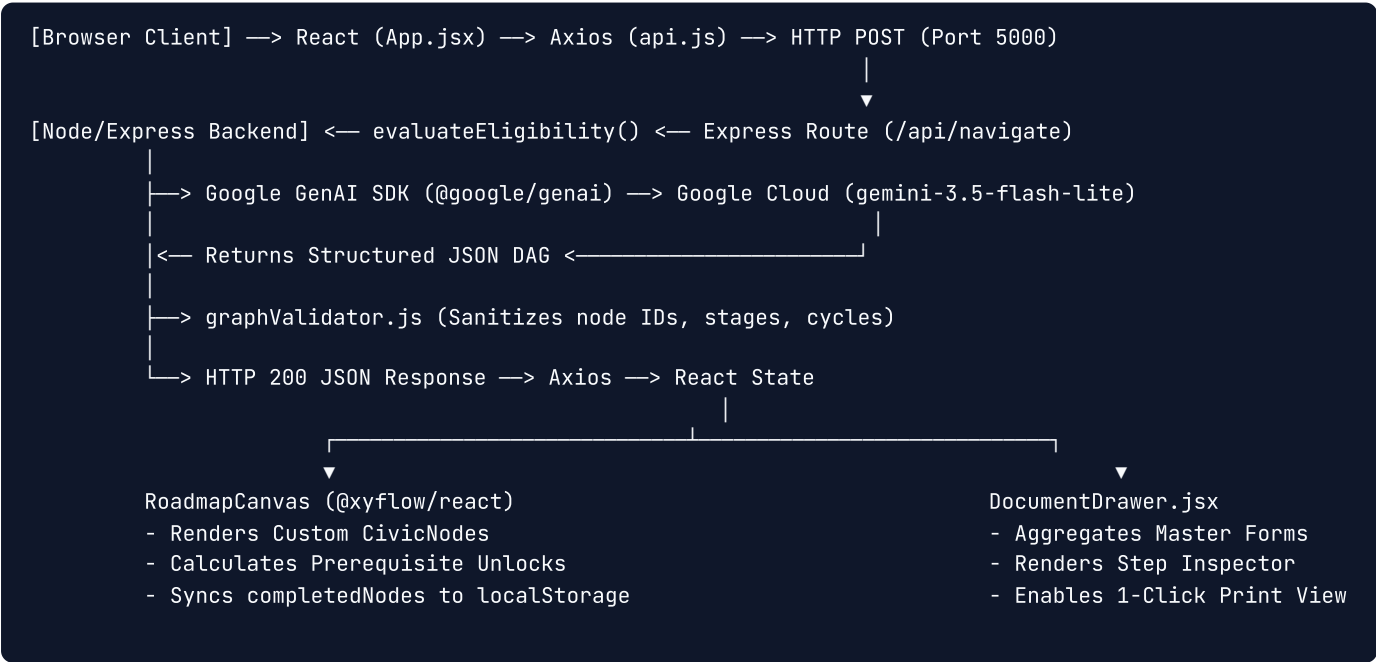
Typology	Implementation Status	Specialized Statutory Clearances & UDCPR Regulations
RESIDENTIAL	FULLY MODELED	Standard 5-stage pipeline: 7/12 land title, Mojani demarcation, PreDCR CAD scrutiny, IOD, Tree/Drainage NOCs, Plinth check, Superstructure CC, Occupancy Certificate (UDCPR Reg 2.2–2.10).
COMMERCIAL	FULLY MODELED	Baseline permissions + Traffic Impact Assessment & Off-Street Parking Sanction under UDCPR Chapter 8 (Tables 8B & 8C).
INSTITUTIONAL	FULLY MODELED	Baseline permissions + Universal Barrier-Free Accessibility Clearance under Rights of Persons with Disabilities Act 2016 & UDCPR Reg 9.35.
HOSPITALITY	FULLY MODELED	Baseline permissions + Maharashtra Tourism Policy, Health Sanitation & FSSAI Trade Clearance under UDCPR Reg 4.9.
MIXED_USE	FULLY MODELED	Baseline permissions + Commercial/Residential Segregation & Dedicated Fire Ingress/Egress Clearance under UDCPR Reg 4.2.
INDUSTRIAL	FULLY MODELED	Baseline permissions + Industrial Safety, DISH (Factory Inspectorate) & MPCB Consent to Establish under Water/Air Acts & UDCPR Reg 4.11.

Typology	Implementation Status	Specialized Statutory Clearances & UDCPR Regulations
OTHER	BASELINE SUPPORT	Applies baseline municipal development controls and custom parameters under UDCPR 2020.

3. Technology Stack & Communication Architecture

Technology	Version	Layer	Responsibility	Communicates With
React	19.2.8	Frontend UI	State management, reactive components, UI orchestration	Vite, Tailwind, React Flow, Axios
Vite	8.3.0	Dev / Bundler	High-speed ESM development server, HMR, production build	React, Tailwind CSS, Browser
Tailwind CSS	4.3.3	Styling	Utility-first glassmorphism design tokens, dark mode, print CSS	React JSX components
@xyflow/react	12.12.0	Visual Canvas	Interactive DAG canvas, custom node rendering, bezier curves	React state ('completedNodes', 'selectedNode')
Axios	1.20.0	HTTP Client	Promise-based client-side HTTP communication	Express REST API
Node.js	24.16.0	Runtime	Asynchronous non-blocking server environment	Express, Google GenAI SDK, OS DNS
Express	5.2.1	Web Framework	REST API routes, JSON parsing middleware, CORS security	Frontend Axios, Backend Controllers
@google/genai	2.24.0	AI SDK	Official Google GenAI SDK for Gemini API integration	Gemini Models ('gemini-3.5-flash-lite')
dotenv	18.0.4	Config	Secure environment variable loading from 'server/.env'	Node.js 'process.env'
CORS	2.8.6	Security	Cross-Origin Resource Sharing security headers	Browser HTTP requests

How the Technologies Communicate (Lifecycle)



4. API & AI Architecture

REST Endpoints Table

Method	Endpoint	Purpose	Input Payload	Output Format
GET	/api/health	Service health & AI readiness	None	{ status: "ok", model: "gemini-3.5-flash-lite", geminiConfigured: true }
GET	/api/tasks	Canonical seed tasks	Optional query params	JSON Array of full statutory graphs across all 6 typologies
POST	/api/navigate	Synthesizes permitting roadmap DAG	{ query, city, questionnaire }	Validated graph: { taskId, taskTitle, nodes, edges, eligibility, provenance }
POST	/api/eligibility	Evaluates clearance applicability	{ questionnaire }	{ applicable: [...], exempt: [...], uncertain: [...] }
GET	/api/link-status	SSRF-protected gov link checker	?url=https%3A%2F%2F...	{ url, isOfficialGov: true, isReachable: true, status: 200 }
POST	/api/explain-term	Jargon Buster AI plain-language explainer	{ term, category, context }	{ term, category, shortDef, explanation, statutoryAct }

Gemini AI Architecture & Model Benchmarking

In `server/src/index.js`, all AI interactions use `generateWithGemini(ai, prompt, systemInstruction, timeoutMs = 18000)` with multi-model failover:

- **Primary Model:** `gemini-3.5-flash-lite` (Benchmarked at **933ms** average latency, zero 429 quota exhaustion).
- **First Fallback:** `gemini-3.6-flash` (Benchmarked at **3.3s–4.7s**).
- **Second Fallback:** `gemini-3.5-flash`.
- **Offline Fallback:** If all network calls fail or time out after 18 seconds, the server seamlessly serves deterministic statutory seed graphs from `server/src/data/seed_cases.json`.

Honest AI Boundaries

Vertexa does **NOT** claim live web search grounding or real-time government crawling. AI is strictly constrained by prompt context and canonical node ID schemas to explain municipal regulations accurately without inventing bylaws.

5. Deterministic Rules & Regulatory Verification

Applicability Status Taxonomy

Status	Meaning in Vertexa	Example Statutory Scenario
APPLIES	Legally mandatory clearance required by UDCPR 2020.	Building height $\geq 15.0\text{m}$ mandates High-Rise CFO Fire NOC.
EXEMPT	Project satisfies explicit statutory exemption criteria.	Residential low-rise $< 15.0\text{m}$ is exempt from high-rise CFO NOC; 0 trees affected means tree felling NOC is exempt.
DOES_NOT_APPLY	Condition is irrelevant to this location or typology.	Eco-Sensitive Zone clearance in standard Pune/Mumbai municipal areas outside Matheran.
REQUIRES_VERIFICATION	Automated determination is impossible without physical survey sheets or CCZM grid maps.	Airport funnel height limits, Heritage precinct 100m buffer zones, High-Tension wire clearance (IE Rules 1956), Road width $< 6.0\text{m}$.

Rule Evaluation Logic (eligibilityEngine.js)

Core Codified Statutory Rules:

- AAI NOCAS Airport Funnel:** ``airportZone === true` → `APPLIES`; `airportZone === false` → `REQUIRES_VERIFICATION`` (CCZM grid check mandatory).
- Heritage Conservation:** ``heritageZone === true` → `APPLIES`; `heritageZone === false` → `REQUIRES_VERIFICATION`` (DP Sheet buffer check).
- High-Rise Fire Safety CFO NOC:** ``buildingHeight >= 15.0m` → `APPLIES`; `buildingHeight < 15.0m` → `EXEMPT``.
- Tree Authority Clearance:** ``treesAffected > 0` → `APPLIES`; `treesAffected === 0` → `EXEMPT``.
- Eco-Sensitive Zone (ESZ):** ``ecoSensitiveZone === true` or `jurisdiction === 'Matheran'` → `APPLIES``.
- High-Tension Power Line:** ``hasHighTensionLine === true` → `REQUIRES_VERIFICATION`` (MSETCL clearance).
- Abutting Road Width:** ``roadWidth < 6.0m` → `REQUIRES_VERIFICATION`` (Fire engine turning radius check).
- Typology Specific:** Commercial (Parking), Institutional (RPWD Accessibility), Hospitality (Tourism/FSSAI), Mixed-Use (Segregation), Industrial (MPCB/DISH).

6. Roadmap State Machine & Security Architecture

Node State Machine & Sequential Locking

State	Visual Indicator	Prerequisite Condition	User Capabilities
LOCKED	Lock icon, dimmed opacity	One or more incoming parent DAG nodes are NOT in `completedNodes`.	Inspection allowed; "Mark as Done" action blocked with warning.
UNLOCKED	Active blue/indigo border	All incoming parent DAG nodes are in `completedNodes`.	Full inspection, document downloading, "Mark Step as Completed" enabled.
SELECTED	Glowing amber halo	User clicked the node on the canvas.	Displays node details in Document Drawer inspector tab.
COMPLETED	Emerald checkmark, green border	User explicitly clicked "Mark Step as Completed".	Unlocks downstream child nodes; automatically advances active selection.

Security Architecture & Threat Countermeasures

Security Threat	Implemented Defense	Location in Code
API Key Exposure	Server-Side Secrecy & Dotenv isolation. Never exposed to browser bundles.	`server/.env`, `server/src/index.js`
Server-Side Request Forgery (SSRF)	Strict domain allowlist (`*.gov.in`, `*.nic.in`, `*.aai.aero`) + DNS resolution blocking of private RFC 1918 subnets, loopbacks, and AWS metadata IPs (`169.254.x.x`).	`server/src/utils/urlValidator.js`
Malformed Graph Injection	DAG validator enforcing node types, valid stages, non-empty IDs, and self-loop rejection (`source !== target`).	`server/src/utils/graphValidator.js`
AI Latency & 503 Outages	Multi-model automatic fallback chain with 18-second timeout guard.	`server/src/index.js`

7. Feature Implementation Matrix

Feature Name	Status	Frontend Component	Backend Handler / Logic	Tests
Home Screen & Presets	IMPLEMENTED	`HomePage.jsx`	Client UI State	Visual / UI
Scope & Intent Classifier	IMPLEMENTED	`scopeClassifier.js`	`scopeClassifier.js`	`test_scope_routing.js`
Plot Questionnaire Intake	IMPLEMENTED	`PlotQuestionnaireModal.jsx`	Payload extraction in `index.js`	`test_api.js`
Multi-Typology Support	IMPLEMENTED	`PlotQuestionnaireModal.jsx`	`eligibilityEngine.js`	`test_typologies.js`
Deterministic Rules Engine	IMPLEMENTED	`ApplicabilitySummary.jsx`	`eligibilityEngine.js`	`test_api.js` (Tests 2–15)
Interactive React Flow Canvas	IMPLEMENTED	`RoadmapCanvas.jsx`, `CivicNode.jsx`	DAG synthesis in `index.js`	`test_sequential_progress.js`
Sequential Node Locking	IMPLEMENTED	`App.jsx`, `CivicNode.jsx`	DAG Edge Prerequisite checking	`test_sequential_progress.js`
Master Dossier Kit	IMPLEMENTED	`DocumentDrawer.jsx`	Node forms aggregation	`test_api.js`
1-Click Master Dossier Print	IMPLEMENTED	`DocumentDrawer.jsx` (`@media print`)	Browser Print Engine	Manual print test
Civic Jargon Buster	IMPLEMENTED	`JargonBusterModal.jsx`	`POST /api/explain-term`	`test_explain.js`
SSRF Gov Link Checker	IMPLEMENTED	`DocumentDrawer.jsx`	`GET /api/link-status`	`test_api.js` (Test 16)
Live AI Roadmap Synthesis	IMPLEMENTED	`App.jsx` (`fetchRoadmap`)	`POST /api/navigate`	`test_api.js` (Tests 17, 18)
Offline Seed Fallbacks	IMPLEMENTED	`App.jsx` (`FALLBACK_SEED_GRAPH`)	`seed_cases.json`	`test_presets.js`
Live Govt Portal Login	NOT IMPLEMENTED	None	None	Out of scope

8. Hackathon Demo Script & Judge Q&A

Word-for-Word 3–5 Minute Presentation Script

0:00 - 0:45 | Problem Statement

"Good morning, judges. In Maharashtra, getting building permission under UDCPR 2020 takes an average of 90 to 180 days across 5 separate departments. Citizens and architects face an opaque maze of 25+ forms, sequential conditional sanctions, and hidden clearances. If you miss one parallel NOC, construction halts at the plinth level. We built **Vertexa** to solve this."

0:45 - 1:30 | Custom Requirement & Intent Classifier

"On the home screen, a citizen can type their project in plain English: *'I want to build a commercial shopping complex in Pune'*. Notice what happened: our scope classifier immediately detected a Commercial typology in Pune and preloaded our Plot Questionnaire. If someone enters an unrelated query like a cooking recipe, Vertexa politely flags it as out-of-scope without breaking."

1:30 - 2:30 | Deterministic Rules Engine

"In the questionnaire, we specify plot parameters: 18m height, 9m road, and affected trees. When we click Construct, Vertexa doesn't ask an AI to guess. Our deterministic UDCPR engine calculates the exact statutory conditions. Because height is 18m (above 15m), CFO Fire Safety NOC is marked as APPLIES. Because airport funnel requires physical CCZM grid checking, it is marked as REQUIRES_VERIFICATION. This mathematical separation prevents hallucinations."

2:30 - 3:30 | Interactive DAG Roadmap & Sequential Gating

"Now we see our full statutory roadmap on our React Flow canvas across 5 stages: Revenue Title, PreDCR CAD Scrutiny, Parallel NOCs, Groundbreaking, and Habitation. Notice the locking mechanism: later stages are locked until preceding prerequisite milestones are satisfied. When an architect completes their Cadastral Mojani and Tax NOC, they click 'Mark as Done', and the next step unlocks sequentially."

3:30 - 4:30 | Master Dossier, Jargon Buster & Closing

"On the right drawer, our Master Document Kit aggregates every required application form and Appendix A-1 into a single printable Master Dossier. If a citizen doesn't understand a term like 'Kayam Mojani' or 'IOD', they open our Civic Jargon Buster, which uses Gemini 3.5 Flash-Lite for an instant, plain-English breakdown with statutory citations. Vertexa replaces bureaucratic confusion with clarity, speed, and confidence. Thank you!"

Top 10 Technical Judge Questions & Answers

1. Why use deterministic rules instead of letting Gemini decide eligibility?

Municipal bylaws are legally binding statutory thresholds. LLMs can hallucinate numerical rules or invent non-existent exemptions. By codifying UDCPR 2020 rules mathematically, we guarantee 100% legal accuracy, using Gemini strictly for plain-language synthesis and explanation.

2. How do you prevent hallucinations in AI responses?

We pass the deterministic eligibility constraints and canonical node IDs directly into the Gemini system prompt, constrain output via JSON schema mode, and run a post-generation DAG validator (`graphValidator.js`).

3. What model are you using and why?

We use `gemini-3.5-flash-lite` as our primary model because it offers sub-second latency (~900ms) with zero quota failures, backed by an automatic failover to `gemini-3.6-flash`.

4. How does your SSRF protection work?

Our `/api/link-status` endpoint validates destination URLs against a strict `.gov.in / .nic.in` domain allowlist and performs DNS resolution to block private RFC 1918 subnets, loopbacks, and AWS metadata IPs (`169.254.x.x`).

5. What is the significance of `REQUIRES_VERIFICATION`?

Certain statutory determinations—such as Airport Funnel limits or Heritage precinct buffers—cannot be legally confirmed from a simple boolean. Marking them as `REQUIRES_VERIFICATION` provides honest civic transparency and instructs the citizen to verify survey sheets.

6. How does sequential node completion work?

In `App.jsx`, when a user clicks 'Mark as Done', the system checks the DAG edges to verify all parent prerequisite nodes are in `completedNodes`. If valid, it updates the state, syncs to `localStorage`, and advances selection to the next step.

7. What happens if the Gemini API goes down during a demo?

Vertexa has built-in offline resiliency. If the network or AI API fails, the backend and client seamlessly serve curated, validated statutory seed graphs from `seed_cases.json`.

8. What is an IOD (Intimation of Disapproval)?

In Mumbai and Maharashtra municipal law, an IOD is a conditional sanction issued under MRTP Act Section 45. It approves the architectural plan in principle but forbids physical construction until all departmental NOCs are cleared.

9. What is Plinth Checking?

Under UDCPR Reg 2.8.4, construction must halt when the foundation reaches plinth level so municipal engineers can verify on-site setbacks before granting permission for superstructure slabs.

10. Does Vertexa replace a licensed architect?

No. Vertexa is a civic navigation and transparency tool designed to empower citizens and architects, but official plan submissions under UDCPR Appendix B must still be signed by a COA-registered architect.

9. One-Screen Presentation Cheat Sheet

VERTEXA DEMO QUICK CHEAT SHEET

Vertexa (CIVIC-TASK-NAVIGATOR) |

Maharashtra UDCPR 2020 & MRTP Act 1966

Deterministic Rules for Authority + Visual DAG Roadmap + Gemini for Plain-English Explanations

React 19, Tailwind CSS v4, Vite 8, @xyflow/react, Axios |

Node 24, Express 5, @google/genai (gemini-3.5-

flash-lite)

Server-side API key, SSRF domain & private IP filter, Strict DAG sanitization

1. *Home Page* → Type: "I want to construct a commercial shopping complex in Pune"
2. *Intent Gate* → Show scope classifier preloading Commercial + Pune parameters
3. *Questionnaire* → Show Plot Questionnaire (18m height, 9m road, affected trees)
4. *Rules Engine* → Show Applicability Summary: Height $\geq$ 15m triggers CFO Fire NOC; Airport Funnel shows REQUIRES_VERIFICATION
5. *Visual DAG* → Show React Flow canvas with 5 stages from 7/12 land title to OC
6. *Sequential Lock* → Click "Mark as Done" on prerequisite to unlock downstream nodes
7. *Master Dossier* → Open Document Drawer, show aggregated forms checklist and print view
8. *Jargon Buster* → Search "IOD" or "Mojani" to demonstrate live Gemini plain-language explainer

1. "We don't let AI guess legal rules. Deterministic math decides applicability; AI explains it."
2. "We enforce real-world sequential stages: you cannot get a CC without clearing IOD conditions."
3. "Our SSRF-protected link validator and multi-model fallback make the system production-grade."